

Relationships Within Triangles

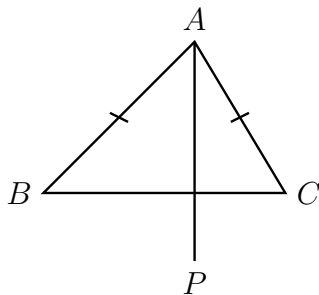
US Common Core – High School Geometry

Domain: Congruence (G-CO.C.10) | Cluster: Prove theorems about triangles

1 Perpendicular Bisectors

Theorem 1 (Perpendicular Bisector, Isosceles Case)

In an isosceles triangle, the perpendicular bisector of the base also passes through the apex, and is the same line as the angle bisector of the apex angle.



Note: Any point E on this perpendicular bisector is equidistant from B and C: $EB = EC$.

Example: Solving for x

$\triangle ABC$ with $AB = 3x + 4$, $AC = 5x + 2$, and AD the perpendicular bisector of BC (also the angle bisector of $\angle A$):

$$5x + 2 = 3x + 4 \Rightarrow x = 1$$

Theorem 2 (Converse of the Perpendicular Bisector Theorem)

If a point lies on a line intersecting a side of a triangle, and the segments joining that point to both endpoints of the side are of equal length, then that line is the perpendicular bisector of the triangle's side.

2 Angle Bisectors

Theorem 3 (Angle Bisector Theorem)

If a point D lies on the bisector of $\angle A$, then D is equidistant from the two sides of $\angle A$.

Theorem 4 (Converse of the Angle Bisector Theorem)

If a point D lies on a line intersecting $\angle A$, and D is equidistant from the two sides of $\angle A$ (i.e. the perpendicular distances match), then that line is the bisector of $\angle A$.

3 Points of Concurrency

Circumcenter

The point where the three perpendicular bisectors of a triangle's sides meet. It is equidistant from the three **vertices**.

Example: Finding a Circumcenter

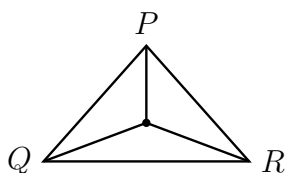
Triangle with $A(2, 6)$, $B(8, 6)$, $C(8, 10)$.

- AB is horizontal; its midpoint is $(5, 6)$, so its perpendicular bisector is the vertical line $x = 5$.
- BC is vertical; its midpoint is $(8, 8)$, so its perpendicular bisector is the horizontal line $y = 8$.

These meet at the circumcenter $(5, 8)$.

Incenter

The point where the three angle bisectors of a triangle meet. It is equidistant from the three **sides**.



Centroid

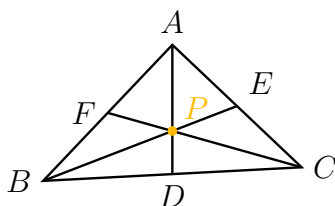
The point where the three **medians** of a triangle meet.

Centroid Theorem

The centroid divides each median in a $2 : 1$ ratio from the vertex:

$$AP = \frac{2}{3}AD, \quad BP = \frac{2}{3}BE, \quad CP = \frac{2}{3}CF$$

where D, E, F are the midpoints of the sides opposite A, B, C .



Altitude and Orthocenter

An **altitude** is the perpendicular distance from a side to the opposite vertex. The **orthocenter** is the point where the three altitudes of a triangle meet.

4 The Midsegment of a Triangle

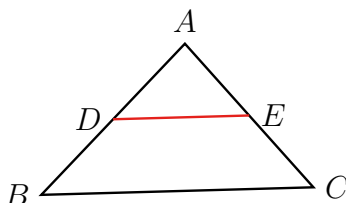
Definition

A **midsegment** connects the midpoints of two sides of a triangle. Every triangle has three midsegments.

Midsegment Theorem

If DE is a midsegment of $\triangle ABC$ (with D, E the midpoints of AB, AC) parallel to BC , then

$$DE \parallel BC \quad \text{and} \quad DE = \frac{1}{2}BC.$$



Coordinate Verification

Triangle $A(-5, 2)$, $B(1, -2)$, $C(-3, 6)$, with midpoints $D(-4, 4)$ of AC , $E(-2, 0)$ of AB , $F(-1, 2)$ of BC .

$$\text{slope of } DE = \frac{0 - 4}{-2 - (-4)} = \frac{-4}{2} = -2, \quad \text{slope of } CB = \frac{-2 - 6}{1 - (-3)} = \frac{-8}{4} = -2$$

Since the slopes match, $DE \parallel CB$ – confirming the Midsegment Theorem for this triangle.

5 The Triangle Inequality

Theorem 1 (Triangle Inequality Theorem)

The sum of the lengths of any two sides of a triangle is always greater than the length of the third side. For $\triangle ABC$:

$$AB + BC > AC, \quad BC + AC > AB, \quad AB + AC > BC.$$

Theorem 2 (Side-Angle Relationship)

If two sides of a triangle are unequal, the longer side lies opposite the larger angle.

Theorem 3 (Converse: Angle-Side Relationship)

If two angles of a triangle are unequal, the larger angle lies opposite the longer side.

Example

In $\triangle ABC$, $\angle B = 50^\circ$ and $\angle C = 30^\circ$. Since $\angle B > \angle C$, the side opposite $\angle B$ is longer than the side opposite $\angle C$:

$$AC > AB.$$

6 Indirect Proof

Idea

In an **indirect proof**, we assume the opposite of what we want to prove, and show that this assumption leads to a contradiction – which means the original statement must be true.

Example: Negating a statement (first step of an indirect proof)

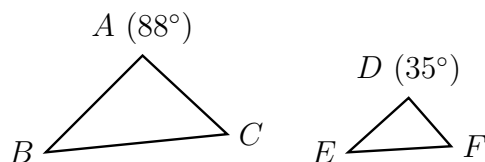
- To indirectly prove “if x and y are even, then xy is even,” write $x = 2n$, $y = 2m$ for integers n, m : then $xy = (2n)(2m) = 4mn$, which cannot be odd.
- To indirectly prove a statement about $\angle A$, begin by assuming its negation, e.g. assume $m\angle A \neq 100^\circ$, or in a different problem, assume M is *not* the midpoint of $\overline{JL}$, so $\overline{JM}$ might be any line not perpendicular to $\overline{KL}$.

Note: The source notes include a worked indirect-proof example (statement: “if $WV + VU \neq 12$ and $VU = 5$, then $WV \neq 7$ ”) whose algebra is difficult to fully verify from the handwriting; the general technique above is reproduced with confidence, but that specific numeric example has been left out rather than risk transcribing it incorrectly.

7 The Hinge Theorem

Theorem (Hinge Theorem / SAS Inequality)

If two sides of one triangle are congruent to two sides of another triangle, and the included angle of the first is larger than the included angle of the second, then the third side of the first triangle is longer than the third side of the second.



Example

$AB \cong DE$, $AC \cong EF$, and $m\angle A = 88^\circ > m\angle E = 35^\circ$. By the Hinge Theorem:

$$BC > DF.$$

Converse (SSS Inequality)

If two sides of one triangle are congruent to two sides of another triangle, and the third side of the first is longer than the third side of the second, then the included angle of the first is larger than the included angle of the second.